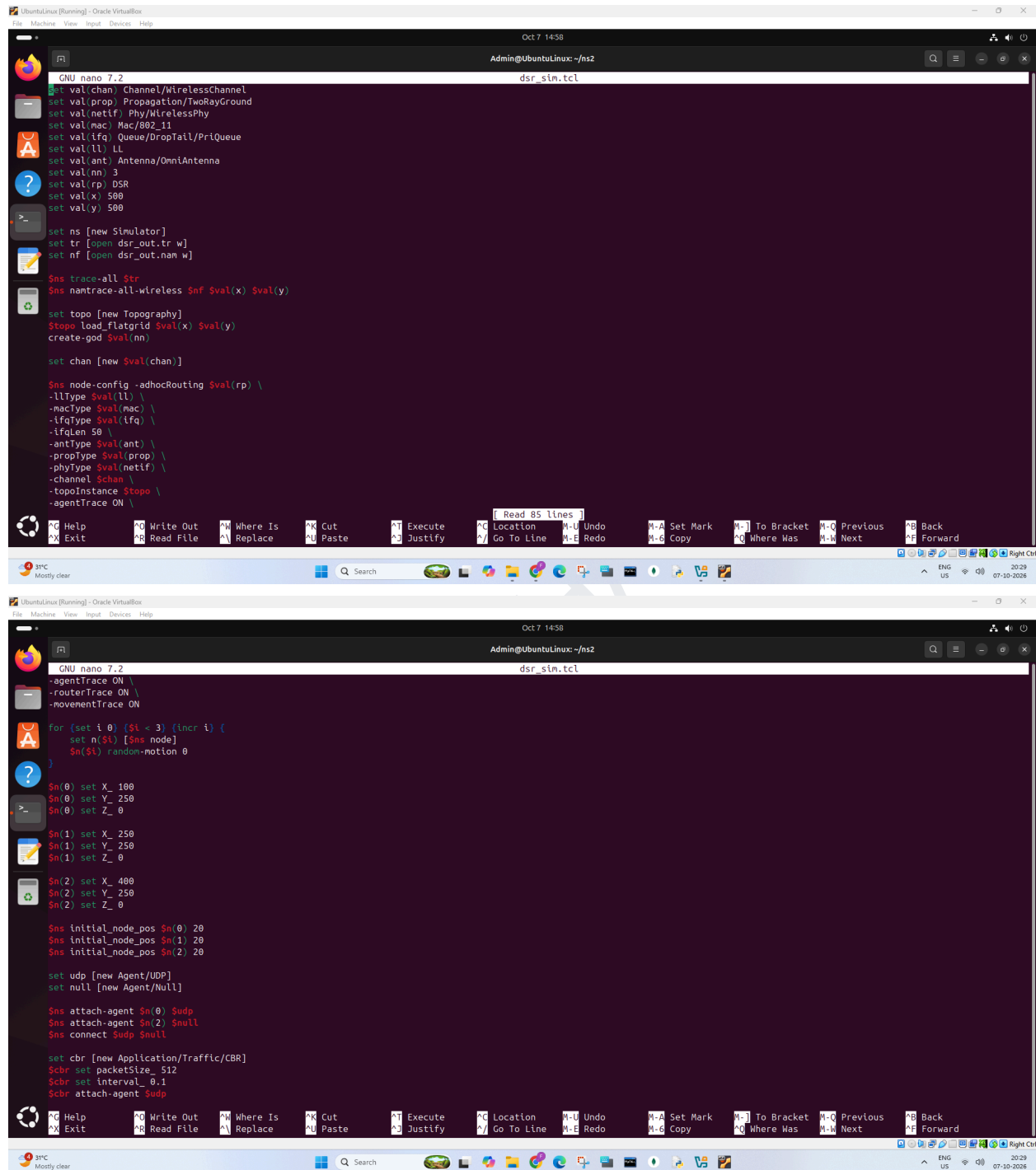


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AIM: Understanding DSR in NS2

TCL Script



The image displays two screenshots of a terminal window running the nano text editor in a virtual machine. The terminal window title is 'Admin@UbuntuLinux: ~/ns2'. The first screenshot shows the configuration of the DSR simulation in the file 'dsr_sim.tcl'. The code includes setting various parameters like channel, propagation, PHY, MAC, queue, antenna, and node configuration. It also sets up the network topology and node movement. The second screenshot shows the execution of the simulation, including setting up the UDP agent, attaching agents, and starting the CBR traffic source.

```
GNU nano 7.2 dsr_sim.tcl
set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmnAntenna
set val(nn) 3
set val(rp) DSR
set val(x) 500
set val(y) 500

set ns [new Simulator]
set tr [open dsr_out.tr w]
set nf [open dsr_out.nam w]

$ns trace-all $tr
$ns namtrace-all-wireless $nf $val(x) $val(y)

set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)
create-god $val(nn)

set chan [new $val(chan)]

$ns node-config -adhocRouting $val(rp) \
  -llType $val(ll) \
  -macType $val(mac) \
  -ifqType $val(ifq) \
  -ifqLen 50 \
  -antType $val(ant) \
  -propType $val(prop) \
  -phyType $val(netif) \
  -channel $chan \
  -topoInstance $topo \
  -agentTrace ON \

for {set i 0} {$i < 3} {incr i} {
  set n($i) [$ns node]
  $n($i) random-motion 0
}

$ns(0) set X_ 100
$ns(0) set Y_ 250
$ns(0) set Z_ 0

$ns(1) set X_ 250
$ns(1) set Y_ 250
$ns(1) set Z_ 0

$ns(2) set X_ 400
$ns(2) set Y_ 250
$ns(2) set Z_ 0

$ns initial_node_pos $ns(0) 20
$ns initial_node_pos $ns(1) 20
$ns initial_node_pos $ns(2) 20

set udp [new Agent/UDP]
set null [new Agent/Null]

$ns attach-agent $ns(0) $udp
$ns attach-agent $ns(2) $null
$ns connect $udp $null

set cbr [new Application/Traffic/CBR]
$cbr set packetSize_ 512
$cbr set interval_ 0.1
$cbr attach-agent $udp
```

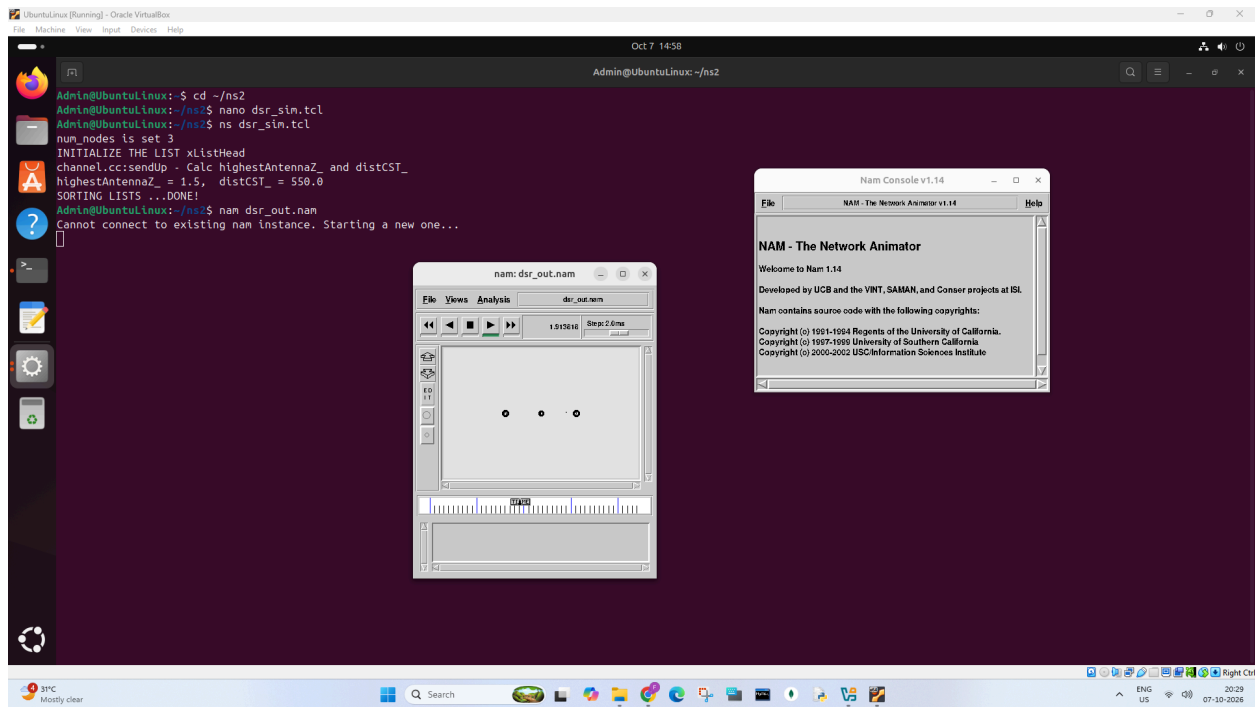
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NS2 Terminal commands & Simulation



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